#Wireshark

#PacketAnalysis

Analyze the ICS-FuelStation compromise using PCAP network traffic analysis.

netcat — no SSH, no web server

to transfer file

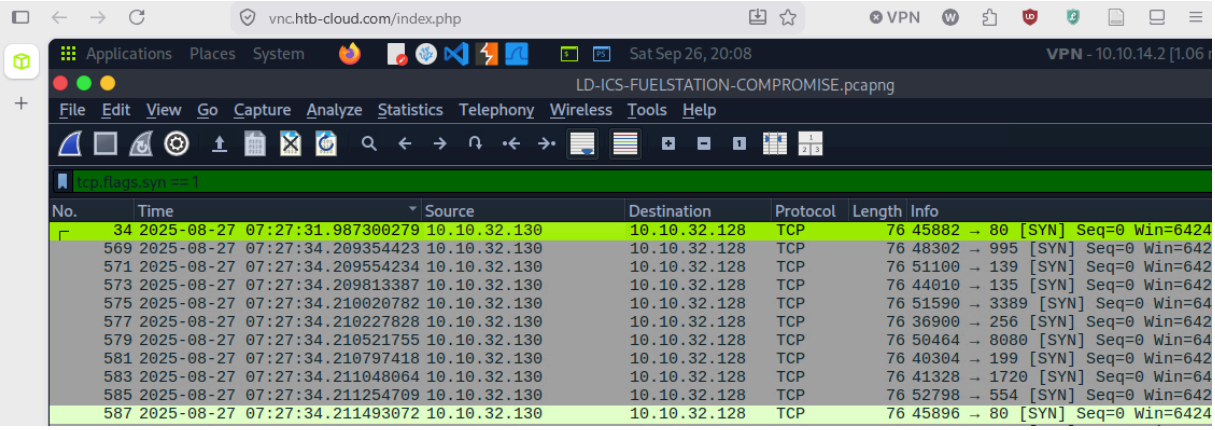
Receiver listens, sender pushes:

```
# on the receiving box
nc -lvnp 9001 > incoming.file

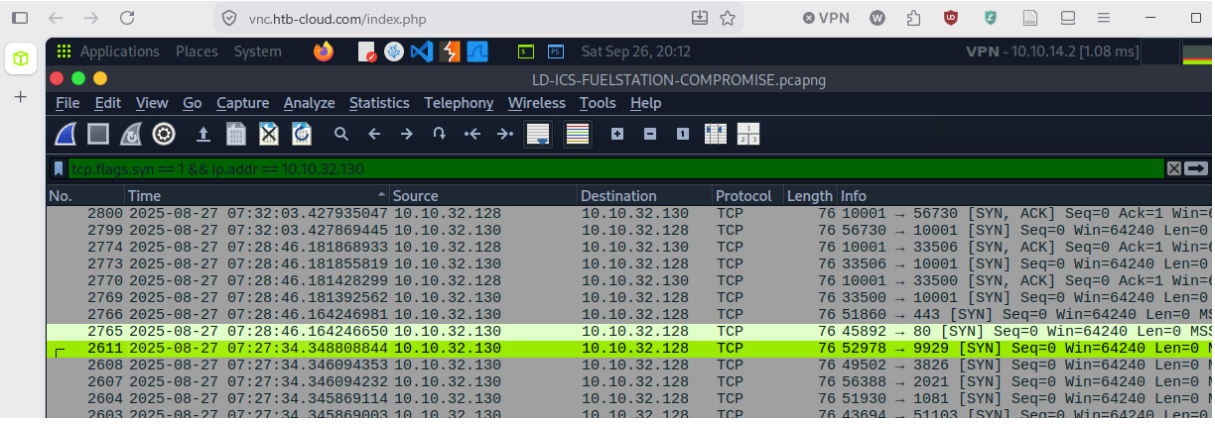

# on the sending box
nc 10.10.10.132 9001 < outgoing.file
```

Q1. The Attacker started off by scanning the subnet. When did port scanning activity start?

Search for flood of TCP with SYN flag packets that indicate port scanning



Q2. When did the port scanning activity stop?



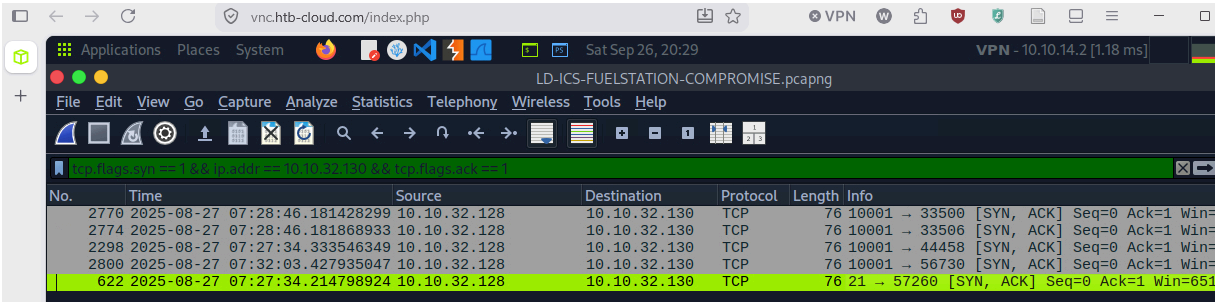
Theres a minute delay between the next TCP packet and the scanning

Q3.Please identify the IP Address of the internal asset used by the threat actor?

10.10.32.130 The source IP of the Scan

Q4. Which port other than the tank gauge system was open on the system?

If TCP is open it would respond with a ACK flag



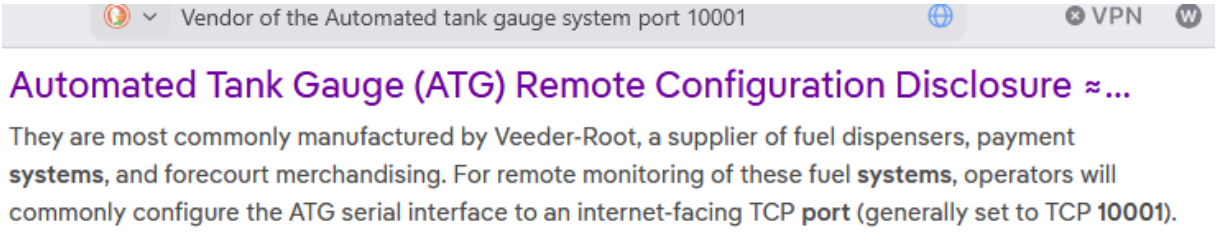
Port 21

Q5. Which port was being used by the ICS system?

Its the other open port

QUESTION 6

Can you confirm the Vendor of the Automated tank gauge system being used on the field?



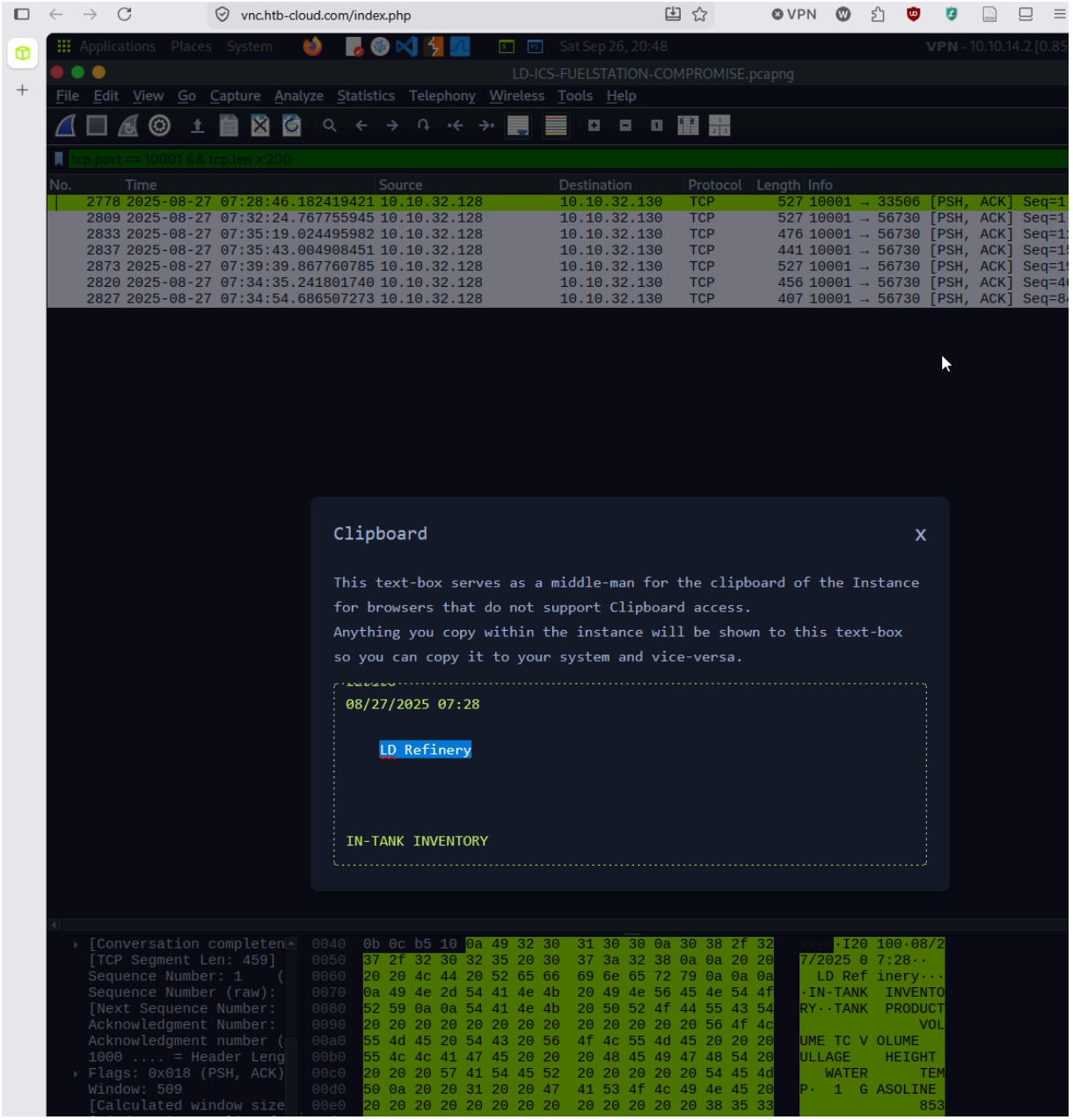
Veeder-Root

Veeder

QUESTION 7

What is the Name of the petroleum pump that was being attacked?

tcp.port == 10001 && tcp.len > 200



QUESTION 8

What is the function code used by the attacker for unauthorized access of data for delivery reports?

I20200

```
8 396.347424389 10.10.32.128          10.10.32.130          TCP        68 10001 → 56730 [ACK] Seq=1 A
9 396.347687583 10.10.32.128          10.10.32.130          TCP       527 10001 → 56730 [PSH, ACK] Se
0 396.347915174 10.10.32.130          10.10.32.128          TCP        68 56730 → 10001 [ACK] Seq=11 
9 526.821478473 10.10.32.130          10.10.32.128          TCP        78 56730 → 10001 [PSH, ACK] Se
0 526.821733378 10.10.32.128          10.10.32.130          TCP       456 10001 → 56730 [PSH, ACK] Se
1 526.821987264 10.10.32.130          10.10.32.128          TCP        68 56730 → 10001 [ACK] Seq=21 
6 546.266192080 10.10.32.130          10.10.32.128          TCP        78 56730 → 10001 [PSH, ACK] Se
7 546.266438911 10.10.32.128          10.10.32.130          TCP       407 10001 → 56730 [PSH, ACK] Se
```

Wireshark · Follow TCP Stream (tcp.stream eq 1005) · LD-ICS-FUELSTATION-COMPROMISE.pcapng

```
IN-TANK INVENTORY

TANK PRODUCT             VOLUME TC VOLUME   ULLAGE   HEIGHT    WATER     TEMP
  1  GASOLINE               8535      8649     6981    82.37     3.96    11.27
  2  FUELOIL                5785      5884     4714    79.39     1.99    -34.66
  3  DIESEL                 14877      14913     3214    69.99     3.42    -23.76
  4  JETFUEL                7939      7964     5586    69.45     2.83    -38.94

^AI20200


I20200
08/27/2025 07:34


LD Refinery


DELIVERY REPORT

T 1:GASOLINE
INCREASE    DATE / TIME              GALLONS TC GALLONS WATER  TEMP DEG F  HEIGHT

      END: 08/27/2025 02:24          8835       8949   3.96      11.27   82.37
    START: 08/27/2025 02:14          8235       8349   3.96      11.27   59.370000000000005
   AMOUNT:                           8535       8649
```

QUESTION 9

At what time was a leak detected from one of the tanks?

```
^AI20300


I20300
08/27/2025 07:34

LD Refinery


TANK 1    GASOLINE
    TEST STATUS: OFF
LEAK DATA NOT AVAILABLE ON THIS TANK


TANK 2    FUELOIL
    TEST STATUS: ON
LEAK DETECTED AT 03:14 AM


TANK 3    DIESEL
    TEST STATUS: ON
HEALTHY


TANK 4    JETFUEL
    TEST STATUS: ON
HEALTHY
```

QUESTION 10

What was the status of the 1s product in the inventory?

```
^AI20500


I20500
08/27/2025 07:35


LD Refinery


TANK    PRODUCT                 STATUS

  1     GASOLINE                                  FILLING-IN-PROGRESS

  2     FUELOIL                                   LOW VOLUME ALARM
                                CONTAMINATION WARNING

  3     DIESEL                                    NORMAL

  4     JETFUEL                                  FULL
```

QUESTION 11

The Attacker got access to the Delivery reports for the station. When was the time for the last delivery?

08/27/2025 02:24

2025-08-27 02:24

```
^AI20200


I20200
08/27/2025 07:34


LD Refinery


DELIVERY REPORT

T 1:GASOLINE
INCREASE   DATE / TIME              GALLONS TC GALLONS WATER  TEMP DEG F  HEIGHT

      END: 08/27/2025 02:24          8835       8949   3.96      11.27   82.37
    START: 08/27/2025 02:14          8235       8349   3.96      11.27   59.370000000000005
   AMOUNT:                           8535       8649
```

QUESTION 12

When did the attacker gain access to the site's shift reports?

2025-08-27 07:35

```
^AI20400


I20400
08/27/2025 07:35

LD Refinery


 SHIFT REPORT

SHIFT 4 TIME: 1:00 AM

TANK PRODUCT

  1  GASOLINE                                VOLUME TC VOLUME  ULLAGE  HEIGHT  WATER   TEMP
SHIFT  1 STARTING VALUES      8535     8649    6981   82.37   3.96    11.27
         ENDING VALUES        9475     9535    7326   135.37  3.96    11.27
         DELIVERY VALUE          1
         TOTALS                7621
```

QUESTION 13

What was the temperature of the 2nd Gasoline product at the time of the incident?

-34.66

```
I20100
08/27/2025 07:39

    LD Refinery


IN-TANK INVENTORY

TANK PRODUCT              VOLUME TC VOLUME   ULLAGE   HEIGHT    WATER     TEMP
  1  YOU_ARE_HACKED          8535      8649     6981    82.37     3.96    11.27
  2  PAY_US_50_ETH           5785      5884     4714    79.39     1.99    -34.66
  3  CONTACT_US_AT          14877     14913     3214    69.99     3.42    -23.76
  4  LEGIONGROUP@PROTONMA    7939      7964     5586    69.45     2.83    -38.94
```

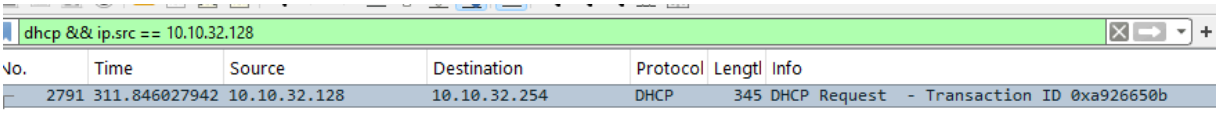
QUESTION 14

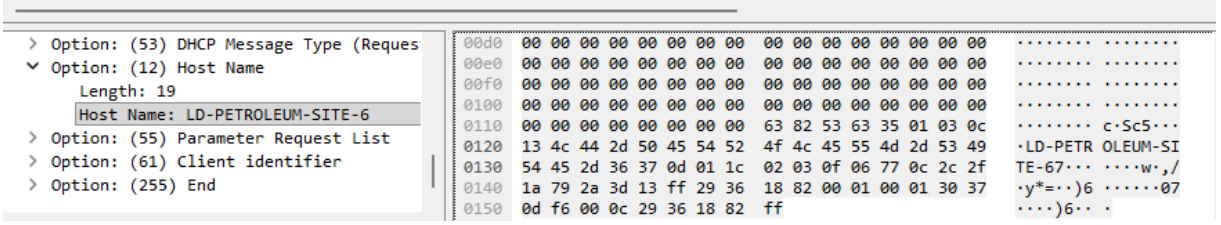
What is the name of the product in the Fourth tank?

JETFUEL From ss in Q9

QUESTION 15

What was the hostname of the machine running the tank gauge software?





QUESTION 16

The Attacker manipulated the tank gauge values and changed the product names to deliver a message to the organization and staff, demanding a ransom payment. What was the ransom amount demanded by the attacker?Answer Format: (2 dogecoin)

50 ETH

50 Ethereum

QUESTION 17

What was the email address associated with the attacker?

LEGIONGROUP@PROTONMAIL.COM